Homework

DM8 - A sequence of integrals - 60 min.

For every $n \in \mathbb{N}$, set $I_n = \int_0^1 x^n e^x \, \mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^x$.

1. Calculate I_0, then I_1 using integration by parts.
2. Prove that, for every $n \in \mathbb{N}$, $I_{n+1} = e - (n+1)I_n$.
3. By comparing the integrands, prove that (I_n) is positive and decreasing.
4. Show that $\frac{1}{n+1} \leq I_n \leq \frac{e}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of (I_n).
5. In the calculation of I_{n+1} by the relation in question 2, an error in I_n is multiplied by $n+1$ if the other rounding errors are neglected? Explain the risk of repeated calculation in this way.